EC7207: High Performance Computing

Group Number: 36

Parallel Protein Secondary Structure Prediction

Parallel Protein Secondary Structure Prediction on CB513 using Serial, OpenMP, POSIX Threads, MPI, and Hybrid (MPI+OpenMP)

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Project Summary

This project addresses protein secondary structure prediction as a residue-level 3-class classification task (H/E/C) on the CB513 benchmark. We will build one common training pipeline and implement it across Serial, OpenMP, POSIX Threads, MPI, and Hybrid (MPI+OpenMP) models to evaluate performance and scalability on available CPU resources.

The dataset contains 513 protein files and 84,119 residues. DSSP labels from the raw CB513 files will be mapped to 3-state labels using a defined rule (H/G/I to H, B/E to E, others to C). Samples will be generated using a sliding window of length 13 with one-hot encoding (20 x 13 = 260 features), ensuring a fair, shared data pipeline for all implementations.

Problem Statement

Serial neural-network training on CB513 is too slow for repeated experiments and parameter sweeps. The key challenge is to parallelize training across shared-memory, distributed-memory, and hybrid settings while keeping prediction behavior consistent with the serial baseline. The project must therefore balance correctness (accuracy parity) and performance (timing/speedup) under real hardware constraints.

Objectives

1. Build a correct serial baseline with reproducible preprocessing, training, and Q3 evaluation on CB513.
2. Implement HPC variants with the same model/data settings: OpenMP, POSIX Threads, MPI, and Hybrid (MPI+OpenMP).
3. Verify correctness by comparing each parallel variant against the serial baseline using Q3 accuracy.
4. Measure runtime, speedup, and scaling by varying threads/ranks/other parameters under fixed hyperparameters.
5. Prepare the final analysis report with a parallelization diagram and implementation description, accuracy comparison against serial, and timing comparison plots for serial vs parallel versions.